Identification of Low Impedance Points Along Railway Tracks From a Railroad Inspection Vehicle

Tiago A. Alvarenga, Rafael A. Nóbrega, Augusto S. Cerqueira, Luciano M. de Andrade Filho, Leonardo M. Honório, Sérgio Rossignoli, Henrique Veloso, and Rafael Batista

***Abstract*— This work addresses the subject of preventive maintenance of railroads which has a considerable impact on the cost and safety of railway companies by proposing a new technique to detect concentrated regions of low impedance between rails along railway lines from a railroad inspection vehicle. Its methodology is based on the standing waves theory applied to transmission lines. Simulation and field measurements have been carried out in order to understand and validate such technique, and a compact measurement system has been designed and implemented producing field-testing results that clearly indicate its feasibility and effectiveness.**

***Index Terms*— Railway inspection, preventive maintenance, low impedance fault location, standing waves theory.**

I. Introduction

RAILWAY traffic systems have been increasingly demanding improvements in predictability and punctuality. Corrective maintenance is one of the factors that have the greatest impact on those requirements. Usually, it involves the interruption of train circulation in the respective rail sections, and traffic restoration occurs only after completion of the corrective operation. Currently, an important part of the railway maintenance structure is based on preventive systems. Predictive maintenance plays a key role in the digital transformation of the railway industry. Many instruments have been developed to collect maintenance-related data to map and predict faults in advance allowing repairs to be properly planned, positively impacting logistics, reliability, safety and costs of train-based transportation systems [1]–[3].

A common problem that generates traffic interruption is the formation of Low Impedance Points (LIPs) across rails [4] concentrated at specific locations on the railway line caused by deposition or gathering of conductive materials over and between rails, making the traffic signaling system to send a warning message to the control system, notifying that such rail sector is occupied and therefore not free for traffic. To overcome the problem, usually a team driving a railroad inspection vehicle is sent to the site to seek for the cause of the false alarm and, depending on its nature, the sector length and its distance from the company installations, the corrective operation may take hours.

With the purpose of reducing the number of corrective interventions due to this kind of false alarm, or the time needed to fix it when the problem is already present, this work proposes a new method to identify LIP across railroad tracks. The proposed technique has been designed to operate in a railroad inspection vehicle [5], [6] knowing that railway companies make use of them not only to perform corrective maintenance but also to travel across the entire railway network with the intention of inspecting the rails conditions so that accidents can be preventively avoided.

The railway track circuit is the most commonly used component for train detection. Railway tracks are divided into sectors with varying length. Each sector is separated from the adjacent ones by an insulating joint so that it can work as a circuit, where current flows from one end to the other. A track circuit [7] is usually designed to continuously detect train presence ensuring, as a fail-safe mechanism, that any cable or bounding failure will also cause the track circuit to be reported as occupied. Every single track circuit is connected to a general traffic management and signaling system responsible for maintaining the operation and the safety of the railway network. One of the simplest track circuits is based on Direct Current (DC) injected into an end of the track while a relay is connected to the other end [8]. The relay state is transmitted to the railway company control system which uses this information to identify train presence. Alternating Current (AC) signaling systems are also used to this purpose, as Audio Frequency Track Circuits (AFTC) [9] which does not require the use of insulating joints. Modern systems combine Communication-Based Train Control (CBTC) and track-circuit technologies for safety enhancement [10].

Transmission Line (TL) theory has been applied for a few decades to describe the electrical characteristics of train tracks [11]–[16] to develop an overall understanding of how AC signals travel along track circuits. However, a general and, at the same time, accurate modeling of it becomes

Manuscript received March 6, 2019; revised July 20, 2019, February 14, 2020, and June 8, 2020; accepted September 16, 2020. Date of publication October 9, 2020; date of current version March 9, 2022. This work was supported in part by Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), in part by Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), and in part by Fundação de Amparo a Pesquisa do Estado de Minas Gerais (FAPEMIG). The Associate Editor for this article was V. Punzo. *(Corresponding author: Rafael A. Nóbrega.)*

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Digital Object Identifier 10.1109/TITS.2020.3026956

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Fig. 1. Railroad inspection vehicle front wheels.

impractical since track characteristics may vary considerably from region to region if the real scenario is considered. Hence, any measurement system applied to railway tracks might be proposed and understood based on the common characteristics of the TL model, but field tests are essential to validate its feasibility. That was the strategy followed by the present work.

In addition to the track circuit, another fundamental component for the maintenance and safety of railways are the railroad inspection vehicles. They are manufactured specifically to be used for corrective and preventive maintenance of railroad tracks. Those vehicles are necessary, for example, whenever the control system identifies a problem on the railroad that must be checked and corrected locally. In case of false track occupancy, the problem shall be quickly fixed in order to safely free the way to the upcoming trains. Railroad inspection vehicles are also used to carry out measurement checks in the search for potential or imminent failures, as a form of prevention [18]. Different types of railroad vehicles are available. The one used in this work is based on a pickup truck having the following characteristics (see Figure 1):

- two pairs of track wheels (retractile), one at the front and another at the back side of the vehicle, what makes it capable of moving along the railway;
- short-circuit between the track wheels to generate an occupancy signal to the control system, for safety reasons;
- a steel brush coupled to each track wheel, to clean up the way and to improve the contact between its wheels and the tracks.

Section II proposes a technique to measure low impedance across rails from a railroad inspection vehicle with the objective of detecting concentrated regions of low impedance along

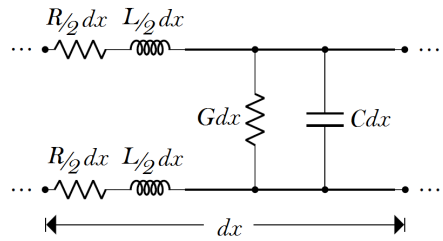

Fig. 2. Distributed element model of a transmission line.

a railway track. The first part of section II presents the theoretical concepts that motivated the development of such technique while the second part is used to validate it and evaluate its characteristics by means of field measurements making use of laboratory instruments. As will become clear throughout the text, the technique validation is completely based on experimental results which have given a clear evidence of its effectiveness. Section III proposes a compact system capable of measuring low impedance values automatically at an appropriate rate and section IV presents the performance of such system based on field tests carried out from a railroad inspection vehicle. Such tests were carried out on rails managed by a Brazilian concessionary named MRS Logística [17] that controls, operates and monitors the Southeastern Federal Railroad Network. All measurements presented in this work were performed on DC-based track circuits separated by insulating joints.

II. Measurement Concept and Validation

This section proposes a technique to identify LIP across rails from a railroad inspection vehicle. It has been divided into two parts: the first one (section II.A) uses a TL-based model to understand how the proposed technique, based on impedance measurement, can behave under conditions that have similarities with the real system; and the second part (section II.B) validates such technique based on field tests making use of laboratory instruments.

A. Impedance Measurement in Transmission Lines

A TL may be characterized by its primary line parameters as indicated in Figure 2 for an infinitesimal length dx, where R represents the resistance per unit length (Ω/km), L the series inductance per unit length (H/km), C the parallel capacitance per unit length (F/km) and G the parallel conductance per unit length (S/km). In special, C and G can vary considerably from one railway to another since they are affected by soil properties and by the rail-to-rail conductance and capacitance, besides being sensitive to climatic conditions.

A TL equivalent circuit based on its characteristic impedance Z_0 is shown in Figure 3. R_S is the source impedance and Z_L the load impedance. The input impedance Z_{in} represents the relationship between voltage and current resulted from the interference between the incident and reflected signals. The source is considered to be at a distance d from the load. A_0 accounts for the voltage amplitude of the sinusoidal injected signal, after the source impedance, and $k = Z_0/(R_S + Z_0)$.

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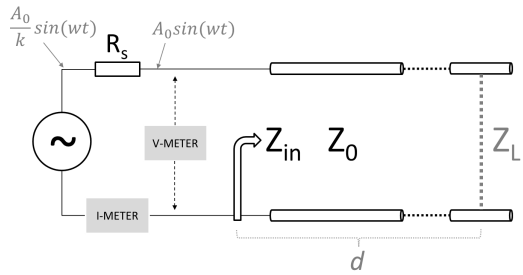

Fig. 3. Transmission line model based on its characteristic impedance Z_0. The equations shown above the R_S terminals consider a sinusoidal signal being injected (only the forward travelling signal component is considered).

The relation between Z_0 and the elements of the equivalent circuit of Figure 2 is given by Equation 1.

$$Z_0 = \sqrt{\frac{R + wL}{G + jwC}} \tag{1}$$

Equations 2 and 3 describe voltage and current components measured at the injection point when a sinusoidal signal is injected into a TL, considering a single reflection point and a signal $A_0 sin(wt)$ being injected just after R_S. A_0 and B_0 ($= A_0/Z_0$) are voltage and current amplitudes of the forward traveling waves at $d = 0$ (at the source side).

$$v(t) = A_0 sin(wt) + A_0 \alpha \Gamma sin(wt + \theta) \tag{2}$$

$$i(t) = B_0 sin(wt) - B_0 \alpha \Gamma sin(wt + \theta) \tag{3}$$

where Γ stands for the reflection coefficient and θ for the phase of the reflected signal, which depends on the injected signal frequency f, on the distance between the injection point and the reflection point d and on the signal propagation speed v, as shown in Equation 4:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \qquad \theta = \frac{2 \cdot d \cdot 2\pi \cdot f}{v} \tag{4}$$

The total signal attenuation depends on the travelled distance and on the TL attenuation constant (α_c) as shown in Equation 5.

$$\alpha = e^{-\alpha_c \cdot 2 \cdot d} \tag{5}$$

Equations 2 and 3 can be understood in terms of Fourier Series as shown in Equations 6 and 7. $2|a_1|$ ($2|b_1|$) models the voltage (current) amplitude of the signal supposed to be readout by a V-METER (I-METER) device positioned as indicated in Figure 3.

$$v(t) = 2|a_1| cos(wt + \angle a_1), \quad a_1 = \frac{A_0}{2j}(1 + \alpha \Gamma e^{j\theta}) \tag{6}$$

$$i(t) = 2|b_1| cos(wt + \angle b_1), \quad b_1 = \frac{B_0}{2j}(1 - \alpha \Gamma e^{j\theta}) \tag{7}$$

An usual alternative model based on Z_{in}, given in Equation 8, is also possible, allowing to replace the TL by a simple impedance body.

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \ tanh(\gamma d)}{Z_0 + Z_L \ tanh(\gamma d)} \tag{8}$$

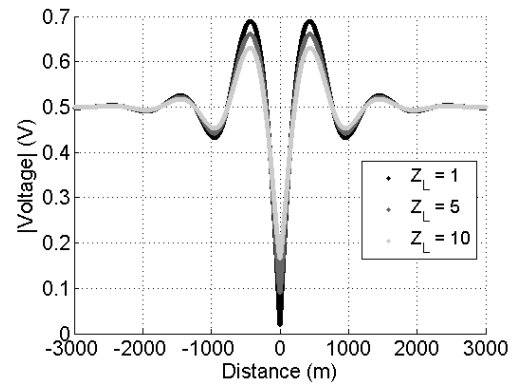

Fig. 4. Distance versus signal voltage amplitude based on the TL model.

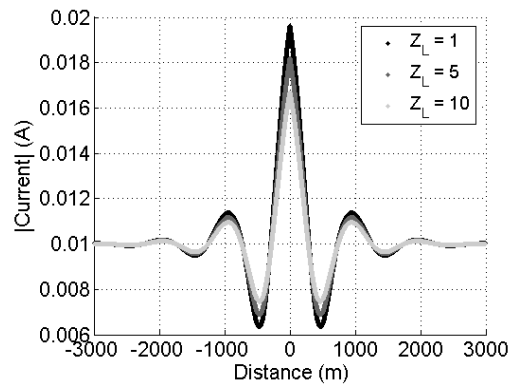

Fig. 5. Distance versus signal current amplitude based on the TL model.

where

$$\gamma = \alpha_c + i\beta \ , \qquad \beta = \frac{\theta}{2d} \tag{9}$$

Figures 4 and 5 show the voltage and current resulting curves (the three alternative models described above provide same results). The following values were used: $v = \frac{2}{3}c$ (c is the light speed), $A_0/k = 1\ V$, $f = 100k\ Hz$, $Z_0 = 50\ \Omega$, $R_S = 50\ \Omega$, $\alpha_c = 0.001$ and $Z_L = 1$, 5 and 10 Ω. Z_L simulates the LIP across the rails and the other parameters were chosen in order not to cancel the reflection effects and, as a consequence, it is possible to see in the resulting curves the periodic component caused by the reflection phenomena. Note that the LIP position is placed at $d = 0$ and negative values of d means that the vehicle is going to the LIP direction.

Impedance might be computed using only the voltage measurement (V_{meas}) or only the current measurement (I_{meas}) as shown in Equation 10.

$$Z = \frac{R_S V_{meas}}{A_0/k - V_{meas}}, \qquad Z = \frac{A_0/k}{I_{meas}} - R_S \tag{10}$$

Figure 6 shows the impedance simulation results. At $d = 0$ all the results converge to the Z_L values as expected.

As specified in Section I, the faced problem is to detect LIPs across rails using a railroad inspection vehicle and those vehicles present short-circuit between their axles, where the injection probes of the measurement system are connected. Therefore, the simulation should consider a low impedance

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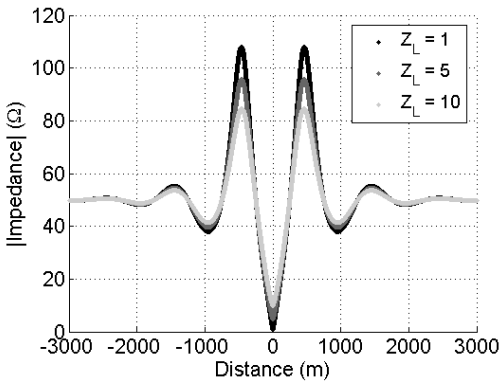

Fig. 6. Impedance measurement based on simulation.

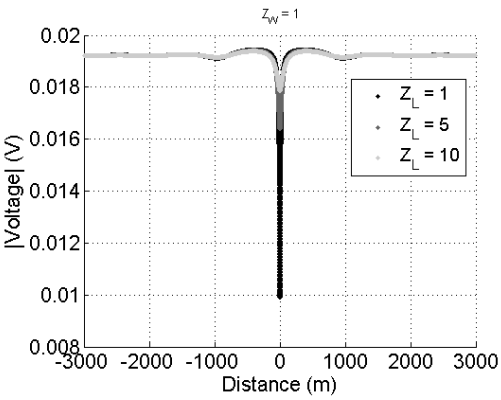

Fig. 7. Distance versus signal voltage amplitude based on the TL model.

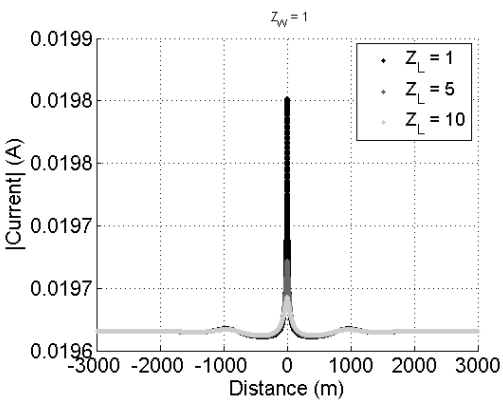

Fig. 8. Distance versus signal current amplitude based on the TL model.

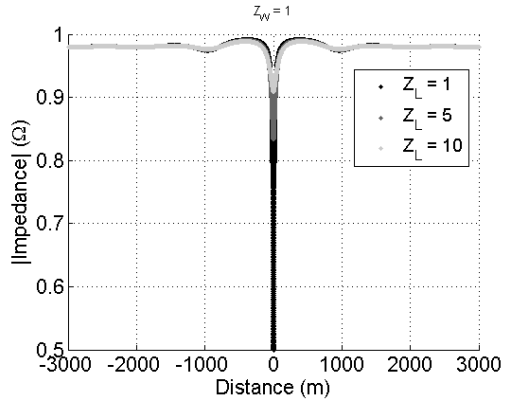

Fig. 9. Impedance measurement based on simulation.

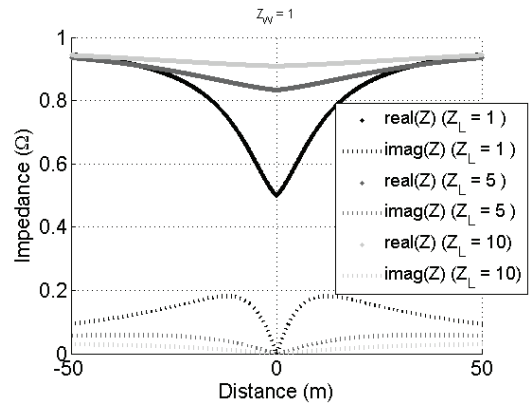

Fig. 10. Real and imaginary parts of the impedance measurement based on simulation for $Z_W = 1$.

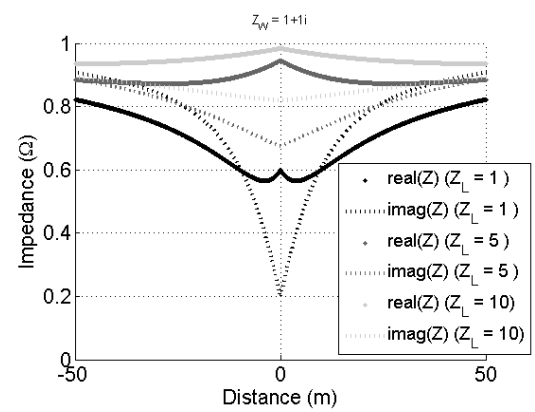

Fig. 11. Real and imaginary parts of the impedance measurement based on simulation for $Z_W = 1 + 1i$.

branch placed in parallel to it (between the vehicle wheels). Figures 7, 8 and 9 show the same voltage, current and impedance plots as before but taking into account such configuration. A resistive impedance (Z_W) of 1 ohm has been considered to simulate this short-circuit between wheels.

At $d = 0$ all the results converge to the value given by the parallel between Z_W and Z_L as expected. As it is possible to observe, the measured magnitudes (voltage, current and impedance) have all been reduced due to Z_W, which would demand a more sensitive measurement system. Additionally, the side lobes caused by the reflection phenomena have all been drastically reduced.

Figures 10 and 11 show the real and imaginary parts of the measured impedance for the case where $Z_W = 1$ and $Z_W = 1 + 1i$, respectively. As it is observed, depending on the value of Z_W, the real or the imaginary part alone might provide a curve with a valley of greater magnitude than the impedance module. Consequently, field tests may be required to define which impedance component will preferably be use,

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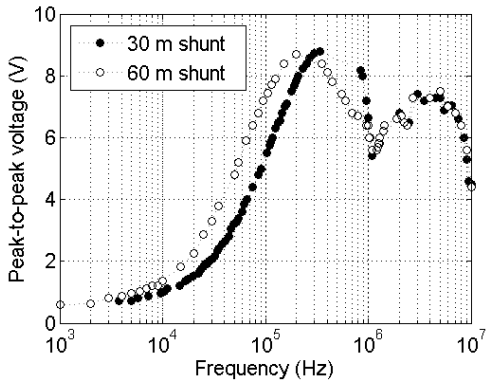

Fig. 12. Frequency sweep for two different shunt positions (30 and 60 meters from the injection point).

although all of them might be continuously monitored. Indeed, field tests have shown that the imaginary part contains the needed information for LIP identification (see Figure 18 for an example). Therefore, the results presented in this work are mainly based on the reactance measurement.

The presented simulation has shown the general waveforms that would be expected when measuring impedance along railway rails using a railroad inspection vehicle and has shown that, depending on the railway and the vehicle characteristics, a high amplitude resolution measurement system might be required in order to make it sensitive enough to LIPs.

B. Validation of the Technique Based on Field Tests

This section is intended to present commercial instruments-based measurements with the objective of validating and delineating the system requirements needed to make it sensitive enough to detect LIPs across railway tracks in the real scenario. It was divided into two parts: while in the first part the frequency responses of a track circuit obtained for two different LIP positions are analyzed, in the second part a situation similar to that found when an inspection vehicle is on the rails is created to enable the evaluation of the behavior of the proposed measurement for short-range measurements.

1) Frequency Scanning: Figure 12 shows the measured frequency response of a standard MRS Logística S.A. railway track section for two shunt positions: at 30 and 60 meters from the injection point. A shunt consists simply of C-clamps and low-resistance wire usually used to simulate the track conditions when a train is passing over it creating a very low resistance connection across the tracks ($< 1\ \Omega$). The result is given in terms of signal frequency versus peak-to-peak voltage and the measurement was performed with a signal generator, with 50 Ω output impedance, set to generate a sinusoidal signal with 5 peak-to-peak Volts of amplitude, having its frequency scanned from 1 kHz to 10 MHz. The signal was injected across the tracks and, in parallel, an oscilloscope was used to measure the signal at the generator output.

As shown, in the low frequency region the resistive component prevails. The inductive component is practically absent at 1 kHz, becoming stronger above a frequency of 10 kHz. In the high frequency region, above 100 kHz, the capacitance component begins to take effect, and the curve behavior becomes irregular. Notice that the closer the measurement device is to the shunt position, the higher is the frequency where the track inductive effect is more sensitive to distance variation. If the intention is to apply a single signal frequency to guarantee a fast measurement rate, what is needed due to the vehicle movement, low distance range and, consequently, higher frequencies are preferred.

Fig. 13. Wooden track cart on ball bearing wheels constructed for the field tests based on laboratory equipment.

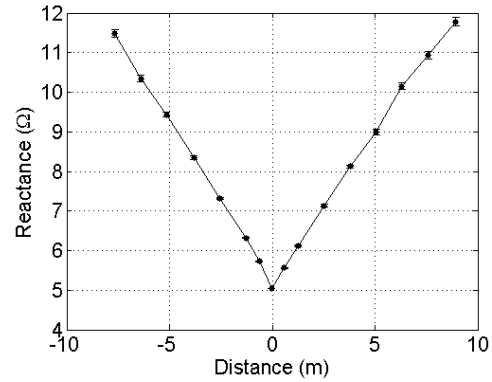

Fig. 14. Impedance measurements using a track cart (no shunt between probes).

2) Short Range Evaluation: For this evaluation it was created a LIP using a shunt at a certain point across the tracks and then a set of impedance measurements was performed around the LIP using a custom-made track cart, equipped with an impedance meter. Its probes were connected to the cart wheels which were in contact with the rails (see Figure 13). Two different configurations were considered: without and with a shunt connected between the cart wheels. For the latter case, a similar cable of that applied across the MRS inspection vehicle wheels was used in order to create a situation as close as possible to the real scenario. The results for both configurations are presented in the following.

a) Without shunt between wheels: In this configuration, measurements were done for a signal frequency of 100 kHz. The results are shown in Figure 14 showing that the LIP valley could be clearly identified, varying from approximately 11.3 Ω to 5.0 Ω in a range of ± 8 meters.

b) With shunt between wheels: In this configuration, measurements were done positioning the shunt in two different locations: between the front wheels (where the measurement probes were connected) and between the rear wheels (two meters far from the measurement probes). The resulting curves are shown in Figure 15. As it is possible to see, the LIP valley amplitude was reduced by a factor ten when compared

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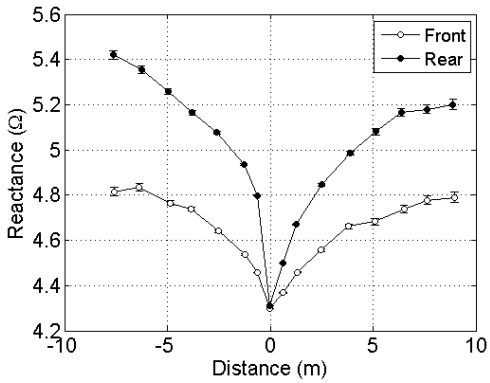

Fig. 15. Measurement tests using a track cart. Shunt two meters far from the device probes (rear axle) and shunt between the device probes (front axle).

to the previous result (Figure 14) due to the cable effect. When the cart shunt is placed two meters away from the probes (changing its position from the front wheels to the rear wheels), the system becomes significantly more sensitive to the LIP, ranging from approximately 5.3 Ω to 4.4 Ω, against 4.8 Ω to 4.3 Ω.

From those tests, other than validating the frequency of 100 kHz to be used for the test signal, it was shown that moving the probes away from the vehicles wheels for a few meters would be enough to improve considerably the measurement system sensitivity. Moreover, for a signal frequency of 100 kHz, the measurement device should have, roughly, an impedance precision better than 0.1 Ω, considering this track cart made in the laboratory. Finally, the measurement rate should be around one per meter to be able to detect LIPs along a railway track. Those values might be more restrictive depending on the measurement noise level in the real scenario and the characteristics of the MRS inspection vehicle (e.g. in addition to the cables connecting its wheels, its axles decrease even more the impedance value between them). Nevertheless, the results given in this section showed the feasibility of the proposed technique and indicated the system upper limits regarding measurement precision and rate, serving as a reference for designing a custom measurement system capable of working coupled to a railway inspection vehicle.

III. System Proposal and Testing

In this section a measurement system implementation is proposed in section III-A. Such system was implemented and coupled to an inspection vehicle for testing. The testing results are described in section III-B. Finally, in section III-C, filtering and detection algorithms are proposed to improve the Signal-to-Noise Ratio (SNR) of the measurements.

A. Hardware Overview

Figure 16 shows a scheme of the proposed measurement system. Its main component is a network analyzer built-in on a chip named AD5933 [19] by *Analog Device* coupled to an additional circuit based on the AD8806 operational amplifier to make it able to measure low impedance values [20].

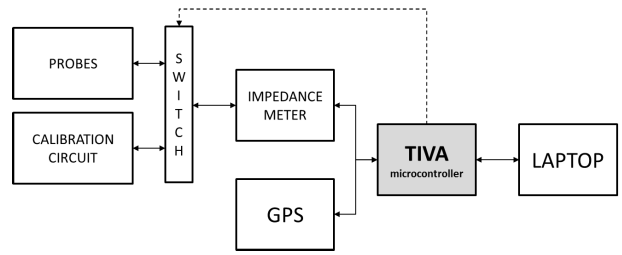

Fig. 16. System block diagram.

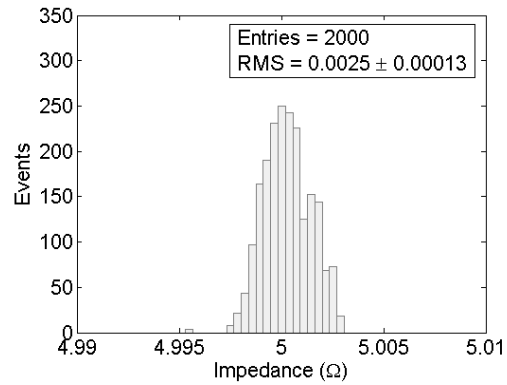

Fig. 17. Precision test of the measurement system.

TABLE I
Measurement Rate Characteristics

Vehicle speed	Meas. per meter	Distance per meas.
60 Km/h	6.7	0.15 m
40 Km/h	10.0	0.10 m
20 Km/h	20.0	0.05 m

The system also includes a Global Positioning System (GPS) to give the location of the detected LIPs, a microcontroller TIVA TM4C123GXL by *Texas Instruments* employed to control the other electronic components and to transfer data from/to a portable computer running a custom C++ based software. Additionally, a decoupling circuit, to be placed between the system signal injection circuit and its probes, and a calibration circuit for the impedance meter have been foreseen.

Repeated measurements with a resistor of 5 Ω have been performed to assess the system precision. The resulting histogram is shown in Figure 17. A system measurement error of 0.0025±0.00013 Ω was estimated.

The system is capable of executing one measurement every 9 ms. Table I shows the measurement rate characteristics considering different railroad inspection vehicle speeds.

B. Railroad Inspection Vehicle Based Measurements

After implementation of the proposed measurement system (described previously in Section III-A), it was possible to study the real signal to be readout by the system. Figure 18 shows the impedance module, real and imaginary parts of a measurement

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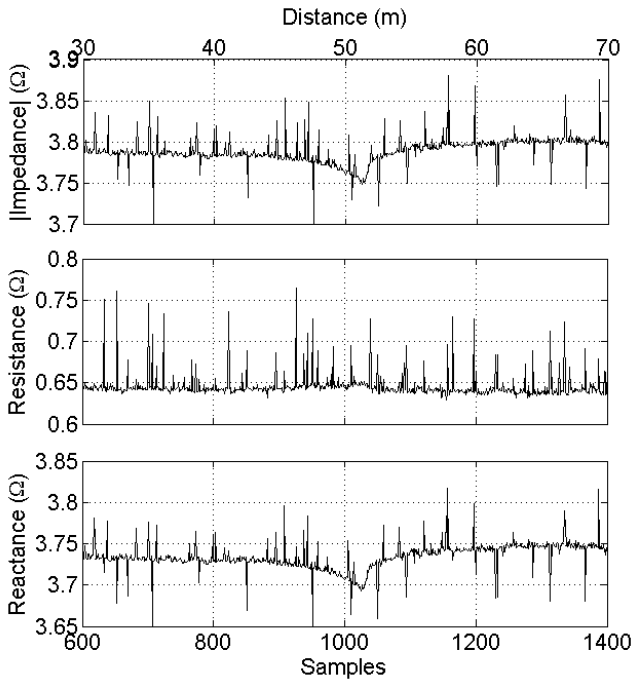

Fig. 18. Impedance module, real and imaginary parts for a measurement carried out from a inspection vehicle at 20 Km/h.

performed with an inspection vehicle at 20 Km/h. The horizontal axis is given in two units: distance (top of the figure) and number of samples (bottom of the figure), both measured from the measurements' starting point. The LIP is located around sample 1025 (or approximately 51 meters away from the measurements' starting point). As can be observed, the relevant information can be obtained by evaluating only the reactance, that being true for all the field measurements taken so far. This possibility was raised by the simulation results presented in section II (see Figure 11). Hence, all the graphs presented in this article focus on the reactance measurement.

Additionally, due to the vibrations experienced by the inspection vehicle as it moves along the tracks, other than a Gaussian noise (see Figure 19), the measurements presented a high level of impulsive noise. To mitigate this impulsive noise and to improve the SNR of the acquired signals, a median filter and a matched filter are employed, as it will be described in Section III-C. As it will be shown, the matched filter design makes use of the noise autocovariance matrix. Figure 20 shows the noise autocovariance function, identifying its non-white nature. As can be seen, the samples are correlated up to approximately the 30th neighboring sample.

C. Signal Processing and LIP Detection

As it was shown in the previous section III-B, when a measurement is carried out in the real scenario, that is using a railroad inspection vehicle traveling at a speed greater than 20 km/h, it generates a very noisy signal and a pre-processing routine is needed in order to make the valley caused by LIPs more easily detectable. Other than the electronics noise, a noise component made of spikes, caused by the constant

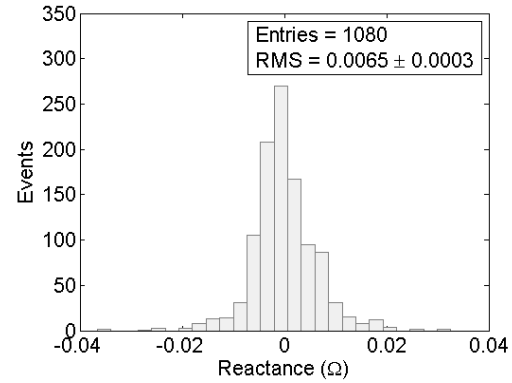

Fig. 19. Noise distribution for the reactance measurement.

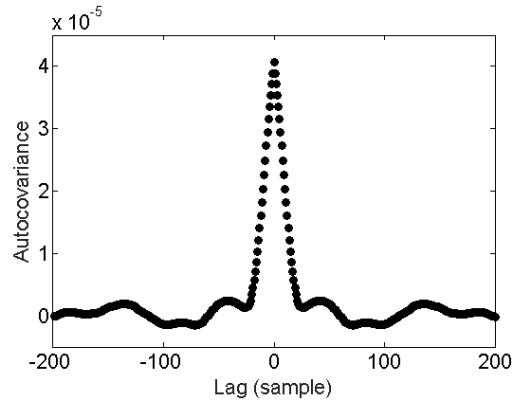

Fig. 20. Autocovariance function of the noise signal after processed by the median filter.

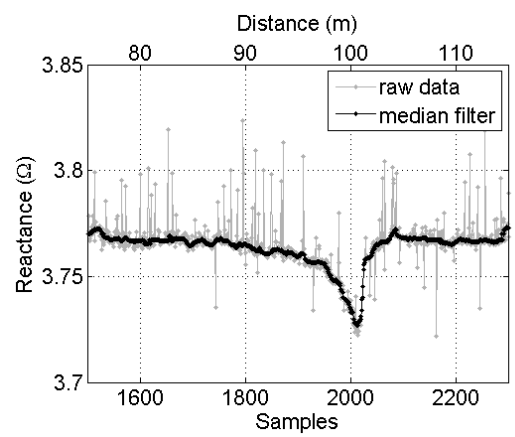

Fig. 21. Example using the median filter for a measurement carried out from a inspection vehicle at 20 Km/h.

vibration of the wheels when running over the rails, is present on the acquired signal. To deal with it a two-step filtering process has been implemented. The first one applies a median filter [21] to the raw signal. Then, its output signal is sent to a matched filter [22] designed to improve even more the SNR and to bring the baseline of the acquired signal to zero.

Figures 21 and 22 show the raw data, the median filter and the matched filter outputs for a measurement performed using the inspection vehicle running at 20 Km/h over rails

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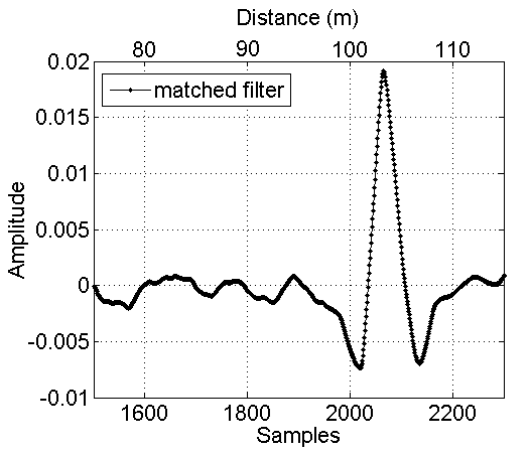

Fig. 22. Example using the matched filter for a measurement carried out from a inspection vehicle at 20 Km/h (same of Figure 21).

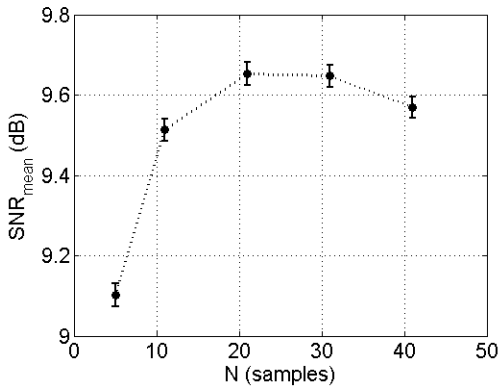

Fig. 23. Number of samples used by the median filter N versus SNR.

with a LIP produced by a shunt cable. As can be observed, the spikes are eliminated by the median filter and the matched filter produces a zero-baseline signal with positive peak. This signal processing procedure makes it straightforward to detect low impedance values by applying a simple threshold value over the matched filter output signal (see Figure 22). Every time a LIP is detected, the operator is warned and the system saves all the related information (signal waveform and GPS coordinates) for future analysis.

The median filter is suitable for mitigating impulsive noise and it consists of replacing the value of each sample $x[n]$ by the median value calculated over a group of neighboring samples $\{x[n-k], \ldots, x[n-1], x[n], x[n+1], \ldots, x[n+k]\}$. The number of samples $N = 2k+1$ to be considered is the only parameter to be defined. Figure 23 shows a graph of N versus SNR. N equal to 21 has been selected to be used during the field measurements.

The matched filter is commonly applied to detect the presence of a known signal in channels with additive noise. The filter operation (in its discrete-time form) in the presence of the signal of interest $\vec{s}$ and noise $\vec{n}$ might be represented as $\vec{y} = \vec{a}^t(\vec{s}+\vec{n}) = \vec{a}^t(A\vec{g}+\vec{n})$ where t stands for the vector transpose operation, $\vec{y}$ for the filter output signal, $\vec{a}$ for the filter coefficients and $\vec{g}$ for the signal of interest normalized

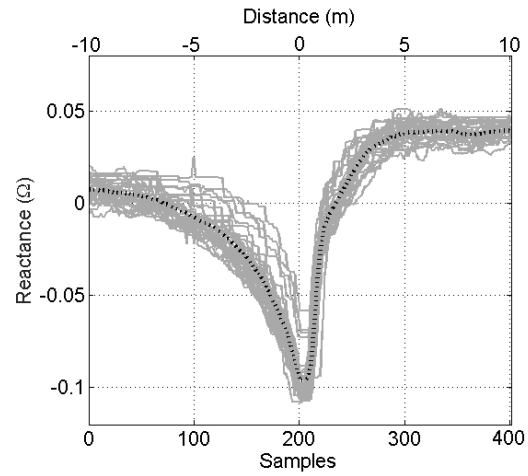

Fig. 24. Signal shape used to implement the matched filter.

by its peak amplitude, given by A. Signal detection is obtained by estimating the amplitude A and verifying if the resulting value is higher than a given threshold. To do so, two constrains should be applied to the optimization problem of finding $\vec{a}$ through minimization of the estimator variance ($\vec{a}^t\mathbf{C}\vec{a}$ where $\mathbf{C}$ is the covariance matrix of the noise signal): one to make the filter output value to be given in terms of peak amplitude (or its absolute value) and another to cancel the signal pedestal variation effect. These two constrains can be mathematically formulated through the following relations, respectively: $\vec{a}^t\vec{g} = 1$ and $\vec{a}^t\vec{1} = 0$. If signal detection is concerned, the former is not mandatory while the latter is important to eliminate the effect of pedestal fluctuation which is certainly an issue to be taken into account, caused mainly by variations in the ballast impedance, which occurs along any railway network, and in the wheel-rail contact impedance. The filter coefficients might be obtained by applying the Lagrange Multiplier technique: The Lagrange function (L) to be minimized is given by Equation 11 [23].

$$L = \vec{a}^t\mathbf{C}\vec{a} - \left(\vec{a}^t\vec{g} - \vec{1}\right) - \vec{a}^t\vec{1} \tag{11}$$

To estimate $\vec{g}$ the following steps were taken: (1) collect several measurements using a shunt connected between rails to obtain a representative sample set; (2) elimination of impulsive noise by using the median filter; (3) temporal synchronization between the acquired LIP signals by minimizing the Mean Squared Error between them; (4) calculation of the average signal; and (5) normalization the resulting signal by its peak amplitude. A plot illustrating step four is shown in Figure 24.

IV. Results

After the implementation of the proposed hardware and the signal processing algorithms, field measurements were applied to assess the performance of the measurement system under two different conditions: one considering a LIP which impedance value is low enough in order to generate a track occupancy alarm signal to the railway control system and another where the LIP impedance is swept from 1.3 Ω to 9.3 Ω; values considered above the alarm condition and thus

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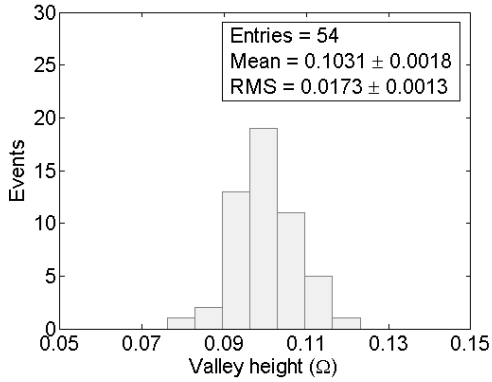

Fig. 25. Distribution of the reactance valley height (obtained with the median filter).

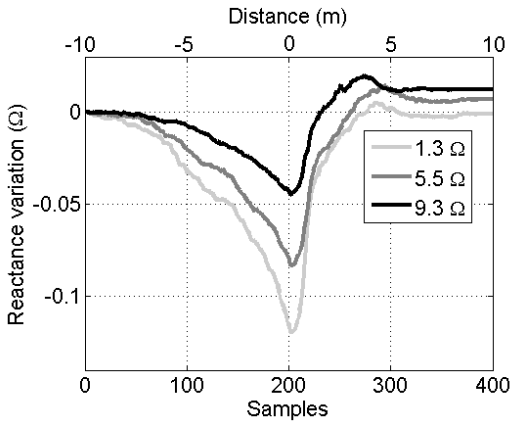

Fig. 26. Waveform examples for different shunt box resistances (after applying the median filter).

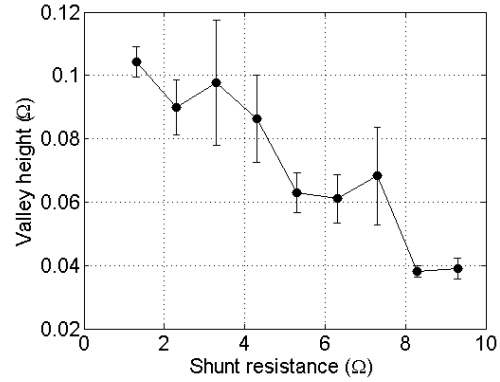

Fig. 27. Shunt resistance versus valley height (obtained with the median filter).

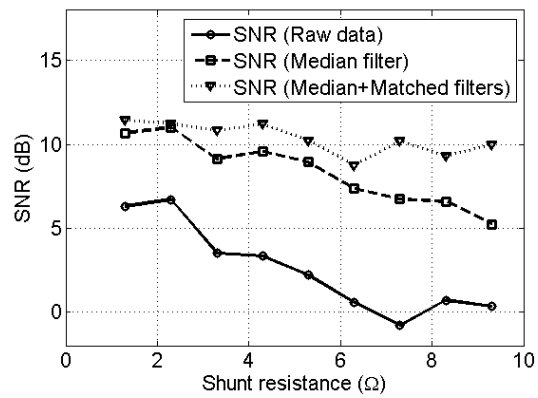

Fig. 28. Shunt resistance versus SNR for the raw data, the median filter and the matched filter.

creating a situation in which a preventive inspection could be put into practice.

A. LIP in Alarm Condition

To work under alarm condition, a shunt was used to connect the rails with a low enough resistance value, roughly below 1 Ω, to force the railway signaling system to generate an alarm, warning that this railroad section is occupied. A set of measurements have been performed in such condition. Figure 25 shows the resulting valley height distribution. In this alarm condition the valley becomes very pronounced, presenting mean and RMS values of 0.1031 ± 0.0018 and 0.0173 ± 0.0013, respectively, well above the noise region. The achieved SNR for the raw data, median filter and matched filter were 6.59, 12.44 and 12.60, respectively.

B. LIP in Prevention Condition

To work in prevention condition, a shunt box, commonly used to simulate track circuit resistances in the testing and commissioning of track feed circuits, was configured to a set of resistance values above 1 Ω, from 1.3 Ω to 9.3 Ω with steps of 1 Ω. For each value, a set of measurements was carried out.

Figure 26 shows the measured waveforms at the median filter output for some of the resistance values (the signal pedestals were canceled to provide a clear view of their differences). It can be observed that the characteristic valley formed around the point of low impedance is being attenuated as a function of the shunt resistance value. Figure 27 shows a graphic of the shunt resistance versus valley height.

Finally, the SNR for each shunt box resistance value is shown in Figure 28, for the raw data, the median filter and the matched filter. As shown, the applied filters practically double the system SNR for low resistance values. For high resistance values, the SNR is almost an order of magnitude higher. In addition, the matched filter tends to provide a higher SNR gain over the median filter for low SNR signals (or, equivalently, high LIP impedance values).

V. Conclusion

This work proposed a new method to preventively find points of low impedance along railway tracks using a railroad inspection vehicle. Simulations were carried out with the intention of understanding the possible results that such method could generate and field measurements using laboratory equipment were performed to validate it in the real scenario. Additionally, a compact measurement system has been proposed, implemented and tested in the field, coupled to an inspection vehicle. The results have clearly indicated that the method works and that it is sensitive to LIPs even for

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values much higher than the alarm trigger level, usually lower than 1 Ω.

ACKNOWLEDGMENT

The authors would like to thank MRS Logística S.A. and the Federal University of Juiz de Fora (UFJF) Postgraduate Program in Electrical Engineering for the support given to this work.

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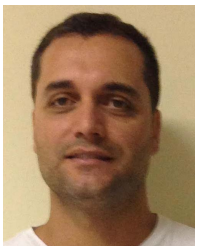

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